IT22026

```
package practicing_multithreading;
import java.util.*;
public class CarParkingSystem {
    static class RegistrarParking {
        private final int carId;
        public RegistrarParking(int carId){
            this.carId = carId;
        }
        public int getCarId() {
            return carId;
        }
        public void process(){
            System.out.println("car " +carId +" is being parked.");
            try {
                Thread.sleep(1000);
            }
            catch (InterruptedException e){
                Thread.currentThread().interrupt();
            }
            System.out.println("car" +carId + "has been parked.");
        }
    }
}
```

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```
static class parkingPool {
    private final Queue<RegistrarParking> orderQueue = new LinkedList<>();
    public synchronized void placeOrder(RegistrarParking order) {
        orderQueue.add(order);
        notifyAll();
    }
    public synchronized RegistrarParking takeOrder() {
        while (orderQueue.isEmpty()) {
            try {
                wait();
            } catch (InterruptException e) {
                Thread.currentThread().interrupt();
                return null;
            }
        }
        return orderQueue.poll();
    }
}
static class ParkingAgent extends Thread {
    private final parkingPool pool;
    private final int agentId;
```

```
private final int totalCars;
private static int completeOrders = 0;

public ParkingAgent(ParkingPool pool, int agentId, int totalCars){
    this.pool = pool;
    this.agentId = agentId;
    this.totalCars = totalCars;
}

@Override
public void run(){
    while(true){
        RegistrarParking order = pool.takeOrder();
        if(order == null) break;
        System.out.println("Agent " + agentId + " is handling car "
            + order.getCarId());
        order.process();
        synchronized(ParkingAgent.class){
            completeOrders++;
            if(completeOrders >= totalCars){
                System.out.println("All cars are parked.");
                break;
            }
        }
    }
}
```

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```
public static void main (String [] args){
    parkingpool pool = new parkingpool ();
    Scanner scanner = new Scanner(System.in);
    System.out.print("Enter number of cars to park: ");
    int numberofcars = scanner.nextInt();
    Scanner.close();
    for (int i=1; i<=3; i++){
        new ParkingAgent(pool, i, numberofcars).start();
    }
    for (int i=1 ; i<=numberofcars ; i++){
        RegistrarParking order =new RegistrarParking(i);
        pool.placeOrder =new RegistrarParking (order);
    try {
        Thread.sleep(500);
    }
    catch (InterruptedException e){
        Thread.currentThread().interrupt();
    }
    }
}
}
```